

# Two Ways to Build an Interval

Unit 3 · Topic 3.10 · THE PROBLEM

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let  $p_O - p_P$  mean Oak minus Pine.

**Elena:** “The two-sample  $z$ -interval is  $(0.030, 0.270)$ .”

**Devon:** “Subtract the farthest endpoints of separate 95% intervals. That gives  $(-0.020, 0.320)$ , so zero is plausible.”

Household samples

| Town | Use | Other | n   | N    |
|------|-----|-------|-----|------|
| Oak  | 77  | 63    | 140 | 2200 |
| Pine | 48  | 72    | 120 | 1800 |

## FIRST TAKE — one sentence before you work the parts

The two reported ranges disagree about whether the towns could be equal. Which range would you trust, and why?

- DOK 1** (a) Define  $p_O$  and  $p_P$  and name the interval procedure for  $p_O - p_P$ .
- DOK 2** (b) Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.
- DOK 3** (c) In 3–4 sentences, choose a method using the independent samples and the variance formula. Explain why subtracting separate 95% intervals does not establish a 95% interval for the difference. Use zero’s location to explain how the method changes the conclusion.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.